AIML Phase 1 — Project Report

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Repository: https://github.com/ashwithagnanasekaran/Growmar_phase1 | Git tag: phase-1

Executive summary

This phase covered data quality auditing, feature engineering, baseline attrition modelling, risk evaluation, and unsupervised attendance-anomaly validation. The work includes reproducible scripts, automated tests, CSV evidence, charts, and written interpretations. Results are experimental and require further independent validation before operational use.

Task completion and evidence

Task	Work completed	Evidence
1. Data audit	Inspected data quality, duplicate employee/date pairs, invalid values, target distribution, and data-quality issues.	data_audit.md, data_audit_issues.png
2. Feature engineering	Built preprocessing and monthly aggregation components; added tests for schema, leakage, and exclusions.	src/features.py, tests/test_features.py
3. Baseline modelling	Compared Dummy, balanced Logistic Regression, and depth-4 Decision Tree using stratified 5-fold cross-validation.	baseline_metrics.csv, benchmark comparison
4. Attrition evaluation	Generated out-of-fold scores, classification metrics, calibration and lift charts, and risk bands.	attrition_evaluation.md, risk_bands.csv, charts
5. Attendance anomalies	Trained Isolation Forest without validation labels and compared model flags with synthetic anomaly types.	attendance_anomaly_validation.md, chart

Task 1 — Data quality

The attrition dataset contains 2,389 rows and 42 columns. There are 2,383 unique employee IDs and six duplicate employee/date pairs. There are 349 recorded leavers (14.61%); the audit also identified 213 invalid WFH values for review. The audit documented issues rather than silently treating questionable values as correct.

Task 2 — Feature pipeline

The implementation separates feature preparation, preprocessing, and monthly attendance/payroll aggregation. The pipeline uses numeric and categorical preprocessing and excludes the target and identified leakage columns. The feature and pipeline test suite passed.

Task 3 — Baseline modelling

Model	ROC-AUC	PR-AUC
Dummy (always stays)	0.500	0.146
Logistic Regression	0.707	0.318
Decision Tree (depth 4)	0.625	0.226

In the reported cross-validation results, Logistic Regression achieved ROC-AUC 0.7068 and PR-AUC 0.3176. The Dummy model provides a no-skill reference, while the depth-4 Decision Tree achieved ROC-AUC 0.6247. The benchmark comparison is saved separately for reproducibility.

Task 4 — Attrition metrics and risk bands

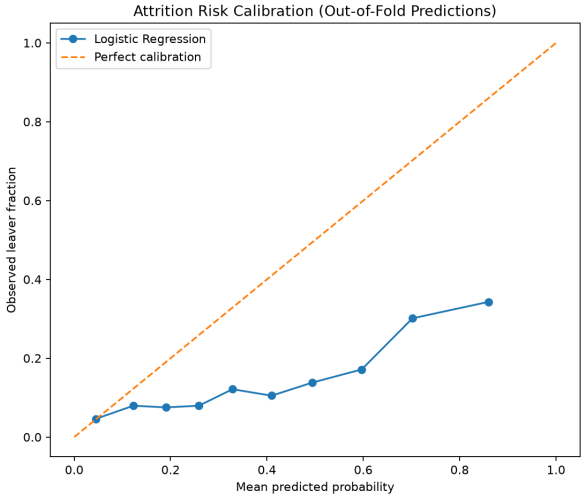
Metric	Result	Interpretation
Accuracy	0.686	Overall classification accuracy
Precision	0.256	Share of predicted leavers who left
Recall	0.599	Share of actual leavers identified
F1	0.358	Balance of precision and recall
ROC-AUC	0.696	Ranking discrimination
PR-AUC	0.291	Precision-recall performance
Brier score	0.208	Probability error; lower is better

These are five-fold out-of-fold probabilities. At the 0.5 threshold, the model achieved recall 0.5989 and precision 0.2555. PR-AUC is useful because leavers are the minority class; accuracy alone can hide poor minority-class detection.

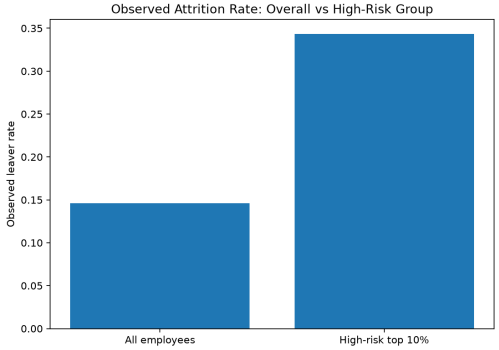
Risk-band analysis

Risk band	Employees	Leavers	Observed leaver rate	Share of leavers captured
High	239	82	34.3%	23.5%
Medium	478	113	23.6%	32.4%
Low	1672	154	9.2%	44.1%

The High-risk band contains 239 employees, of whom 82 were actual leavers (34.31%). This is about 2.35 times the overall historical leaver rate of 14.61%. However, 154 actual leavers were in the Low band, so these bands should not be interpreted as guarantees.



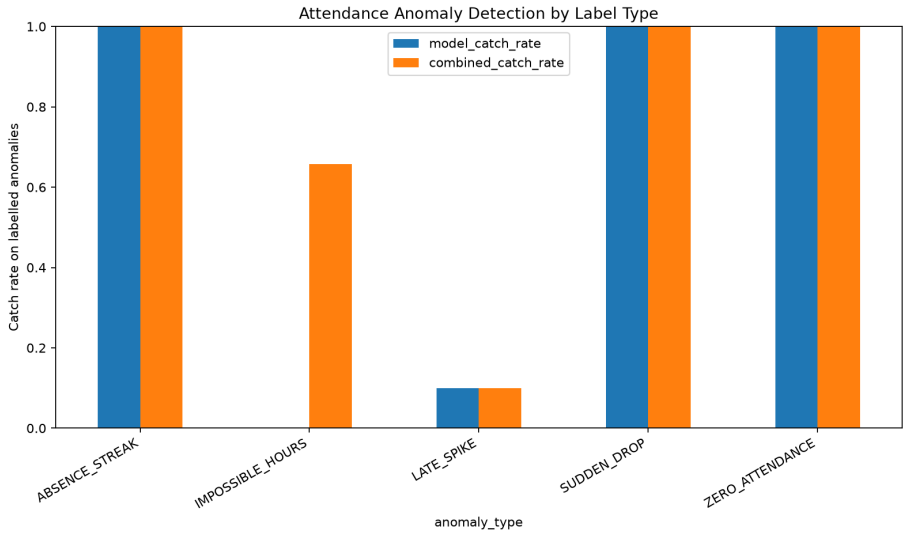
Calibration interpretation: points nearer the diagonal indicate closer agreement between predicted probabilities and observed rates. The chart should be interpreted cautiously because of finite sample sizes.



Task 5 — Attendance anomaly validation

Method	Caught	Missed	Catch rate
Isolation Forest only	88	62	58.7%
Isolation Forest + rules	111	39	74.0%

On 150 labelled anomaly examples, Isolation Forest caught 88 (58.67%) and missed 62. Adding simple attendance rules caught 111 (74.00%) and missed 39. The rules added 23 detections. The strongest remaining weakness was late-arrival spikes: only 3 of 30 were caught by either method. The type-level breakdown is available in the CSV evidence.



Testing and reproducibility

The automated test suite completed with 5 passed tests. It covers feature engineering, model-ready output, target/leakage exclusion, pipeline save/load, and the error raised when transforming before fit. The project contains separate scripts and CSV/Markdown evidence for re-running the core evaluations.

Limitations and responsible use

The attendance validation sheet contains labelled anomaly examples but not a matched set of confirmed normal records. Therefore, this evaluation measures catch rate on the supplied labelled examples but cannot establish precision or false-alarm rate. The workbook's baseline precision/recall values are reference values, not reproduced results. Attrition scores are estimates, not proof of future behaviour. They should support voluntary, human-reviewed support discussions, never stand-alone disciplinary or employment decisions. Further validation on independently reviewed data and fairness checks are needed.

Learning and next steps

This phase strengthened skills in data auditing, feature preprocessing, cross-validation, probability-based evaluation, risk ranking, anomaly detection, testing, and evidence documentation. Next steps are to improve late-arrival anomaly rules, obtain a labelled normal validation set, assess calibration and fairness on future data, and package the pipeline for repeatable deployment.